

ANDRÉ GAMA DE MATOS

ELECTRONICS TEST & R&D ENGINEER | EMBEDDED SYSTEMS | TEST AUTOMATION | COMPUTER VISION
& INDUSTRIAL AI

Manaus, Amazonas, Brazil | andrematos37@hotmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Electronics Test and R&D Engineer with 26 years of experience in the electronics industry, spanning Product Engineering, Test Engineering, NPI, Reliability, Failure Analysis and technology development. Experienced in designing and deploying industrial computer vision, test automation and IIoT solutions that combine Python, OpenCV, TensorFlow/Keras, embedded systems, sensors, microcontrollers and production traceability. Current Resident Researcher in Microelectronics at CBPF, expanding expertise in SystemVerilog, RTL, verification, communication protocols, Cadence tools and PDK-based integrated-circuit workflows. Fluent English for technical communication and global collaboration.

CORE EXPERTISE

Electronics Test / PCBA: ICT/FCT, functional test, fixtures and test programs, troubleshooting, CPK, FMEA, reliability and NPI.

Computer Vision / AI: OpenCV, TensorFlow/Keras, CNNs, MobileNetV2, ROI processing, datasets, model validation and deployment.

Product / Manufacturing: Hardware and firmware validation, BIOS, drivers, Windows/Linux images, SAP/BOM, ECO/ECR, servers and databases.

Embedded / Automation: Python, C/C++, ESP32, Arduino, industrial sensors, serial communication, IoT/IIoT and web interfaces.

Digital Design / Microelectronics: SystemVerilog, RTL, testbenches, SPI, ASIC flow, Cadence Xcelium, synthesis, timing and PDK workflows.

R&D / Documentation: Proofs of concept, experimental validation, root-cause analysis, technical reports, manuals, releases and technology transfer.

SELECTED PROJECTS

SVC USB - Multi-product automated connector inspection

Designed and deployed a production-line computer vision system for USB-A and USB-C connectors, with model-specific recipes, Deep Learning classification, QR-triggered inspection, operator/engineering interfaces, traceability, yield reporting and industrial validation. Registered with INPI. [GitHub](#) | [DOI](#)

SVC Springs - Dual spring inspection platform

Developed an end-to-end inspection system using Python, OpenCV, TensorFlow/Keras, MobileNetV2, Streamlit, USB camera, industrial sensor and Arduino. Implemented dual-ROI inspection, operational dashboards and production reports; validated with more than 1,000 production units. Master's research and INPI-registered software. [GitHub](#) | [DOI](#)

SAPA IoT - ESP32-based industrial automation platform

Designed and deployed an embedded IIoT system for autonomous power control and energy management of industrial audio equipment, including Wi-Fi, multi-window scheduling, manual/automatic modes, web administration, status logging and field installation. [GitHub](#) | [DOI](#)

SPI Master Controller - Private microelectronics residency project

Developed and verified an 8-bit full-duplex SPI Master in SystemVerilog, including RTL, testbench and simulation in Cadence Xcelium. Source code remains private due to residency and confidentiality obligations.

PROFESSIONAL EXPERIENCE

Brazilian Center for Physics Research (CBPF) - Resident Researcher in Microelectronics Jul 2026 - Present

Remote / Manaus, Brazil

- Research and practical development in digital circuit design and verification using SystemVerilog, RTL, testbenches, communication protocols, Cadence Xcelium and PDK-based workflows.
- Developed and verified an 8-bit SPI Master controller and conducted a structured technical review of SPI communication in imaging systems.

Salcomp PLC - Senior Product Engineering Analyst

Mar 2023 - Jul 2026

Manaus, Brazil

- Product Engineering, Test Engineering, Reliability and R&D for smart chargers and switching power supplies used in smartphones, tablets, notebooks and POS terminals.
- Developed test programs and fixtures, supported functional testing and NPI validation, performed circuit and failure analysis, reliability investigation, CPK evaluation and troubleshooting.
- Designed and deployed Industrial AI and IIoT solutions integrating cameras, sensors, Arduino/ESP32, software, traceability and production dashboards; collaborated with global R&D teams and authored technical reports, validation documents and installation manuals.

Positivo Tecnologia - Senior Product Analyst

Sep 2022 - Dec 2022

Manaus, Brazil

- Product, Test and Technology Validation Engineering for notebooks, desktops, tablets, mobile phones and POS terminals; technically coordinated validation and automation activities.
- Developed and maintained test scripts, BIOS, drivers, Windows/Linux images, operating systems, IMEI keys, server infrastructure and technical documentation.

Western Digital - Engineering Specialist - NPI & Technology Validation

Nov 2011 - Jun 2022

Manaus, Brazil

- Led Product Engineering and NPI activities for HDD products, including software, firmware, test-system and industrial-infrastructure validation.
- Supported servers, databases and process-control systems, and worked with global teams in the United States and Asia on product introduction, reliability and failure reduction.

Digitron - Product Engineering Analyst

Oct 2010 - Nov 2011

Manaus, Brazil

- Product Engineering and NPI for HDDs, including test-system validation, failure analysis, firmware/software updates and manufacturing support.

Samsung Electronics - Process Engineering Technician (Junior / Mid-Level / Senior)

Jul 2004 - Oct 2010

Manaus, Brazil

- Product, Process and NPI Engineering for TVs, DVD systems, audio products, MP3 players and monitors; hardware/firmware validation, prototype analysis and test-equipment updates.
- Managed BOM/SAP and ECO/ECR implementation and collaborated with global R&D in South Korea on product introduction, circuit analysis and technical problem solving.

Panasonic - Quality Technician / Product Validation

Feb 2001 - Jun 2004

Manaus, Brazil

- Product validation and Quality Engineering for TVs, including functional testing, embedded hardware/software validation, sample analysis and technical documentation.

Elsys - Electronics Technician Intern

Jan 2000 - Oct 2000

Manaus, Brazil

- Diagnosed, validated and repaired pay-TV decoders and electronic boards using oscilloscopes, spectrum/audio analyzers, multimeters, frequency counters and adjustable AC sources.

EDUCATION & SPECIALIZED TRAINING

Professional M.Sc. in Electrical Engineering - Embedded Systems, UEA | 2024 - 2026, in progress

B.Sc. in Control and Automation Engineering, UniFatecie | 2023 - 2026, in progress

B.Sc. in Software Engineering, UNIASSELVI | Completed 2025

Postgraduate Specialization in Automated Testing for Intelligent Systems, UEA | 2024

Postgraduate Specialization in Software Engineering and Architecture, PUC Minas | 2023

Postgraduate Specialization in Electronics and Computer Engineering, Unyleya | 2023

Technology Degree in Industrial Automation, UNIP | 2014

Electronics Technician Diploma, IFAM | 2000

Microelectronics Training Program, CI Amazonia / iRede | 320 hours, completed Jul 2026, final grade 9.60

Additional Training: FMEA - Failure Mode and Effects Analysis | Metallography | Advanced SMT for Industry 4.0 | AATEC DAY 2025 - Artificial Intelligence.

LANGUAGES & INTELLECTUAL PROPERTY

Portuguese: Native | English: Fluent | Spanish: Limited professional proficiency.

INPI software registrations: SVC Springs - BR512026003433-7 | SVC USB - BR512026005765-5.